

API Testing Report

Name : Shahbaz Ahmad

GitHub branch: capstone/307424-shahbaz-ahmad

Date: 21-07-2026

Section C : - Api Automation

Objective

To design, develop, and execute API automation tests ensuring reliability, scalability, and correctness of REST APIs using a structured framework. Website <https://demoqa.com/>

GithubLink:- [RajaVUst/SDET-2 at capstone/307424-shahbaz-ahmad](#)

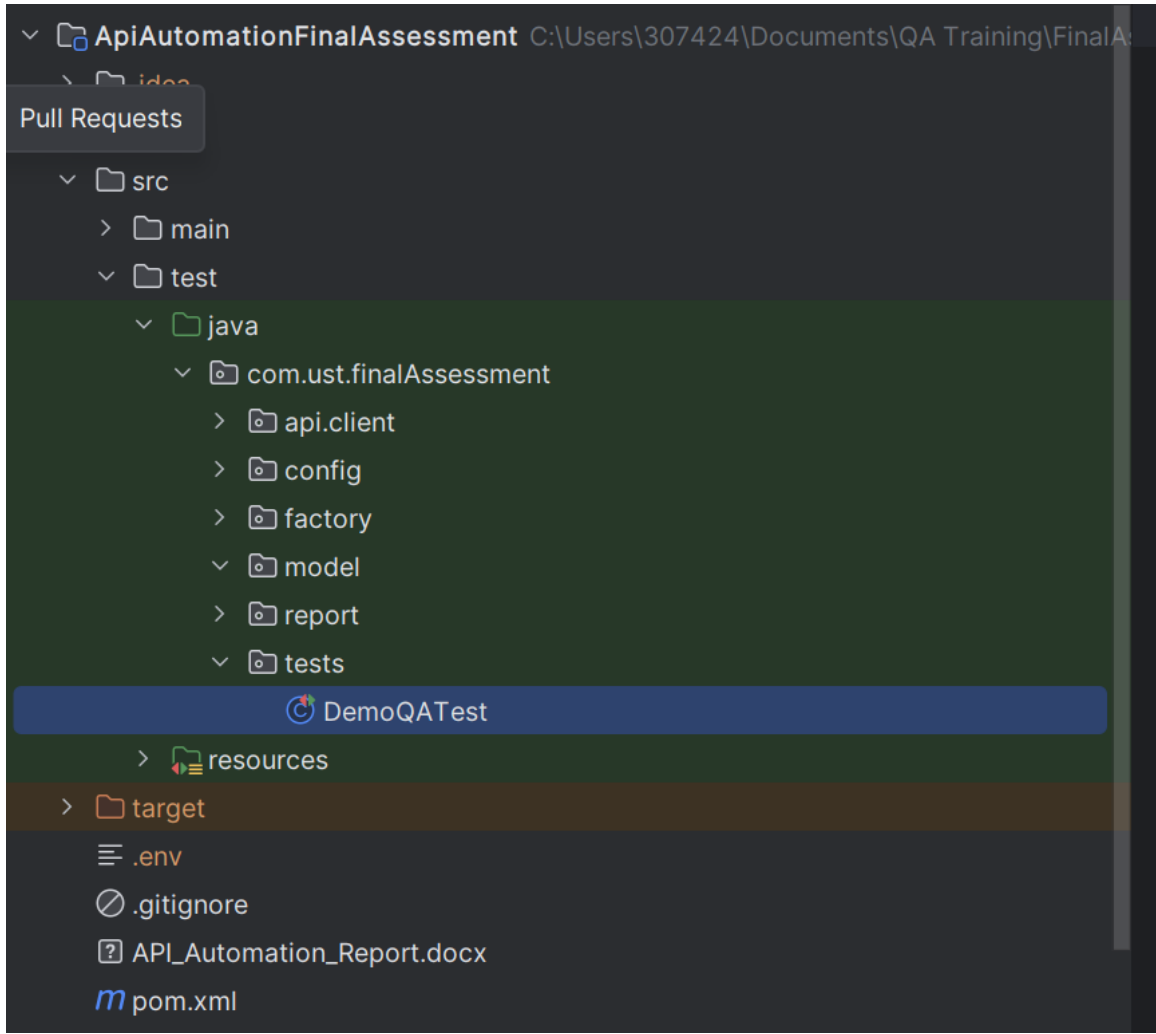
Environment Setup

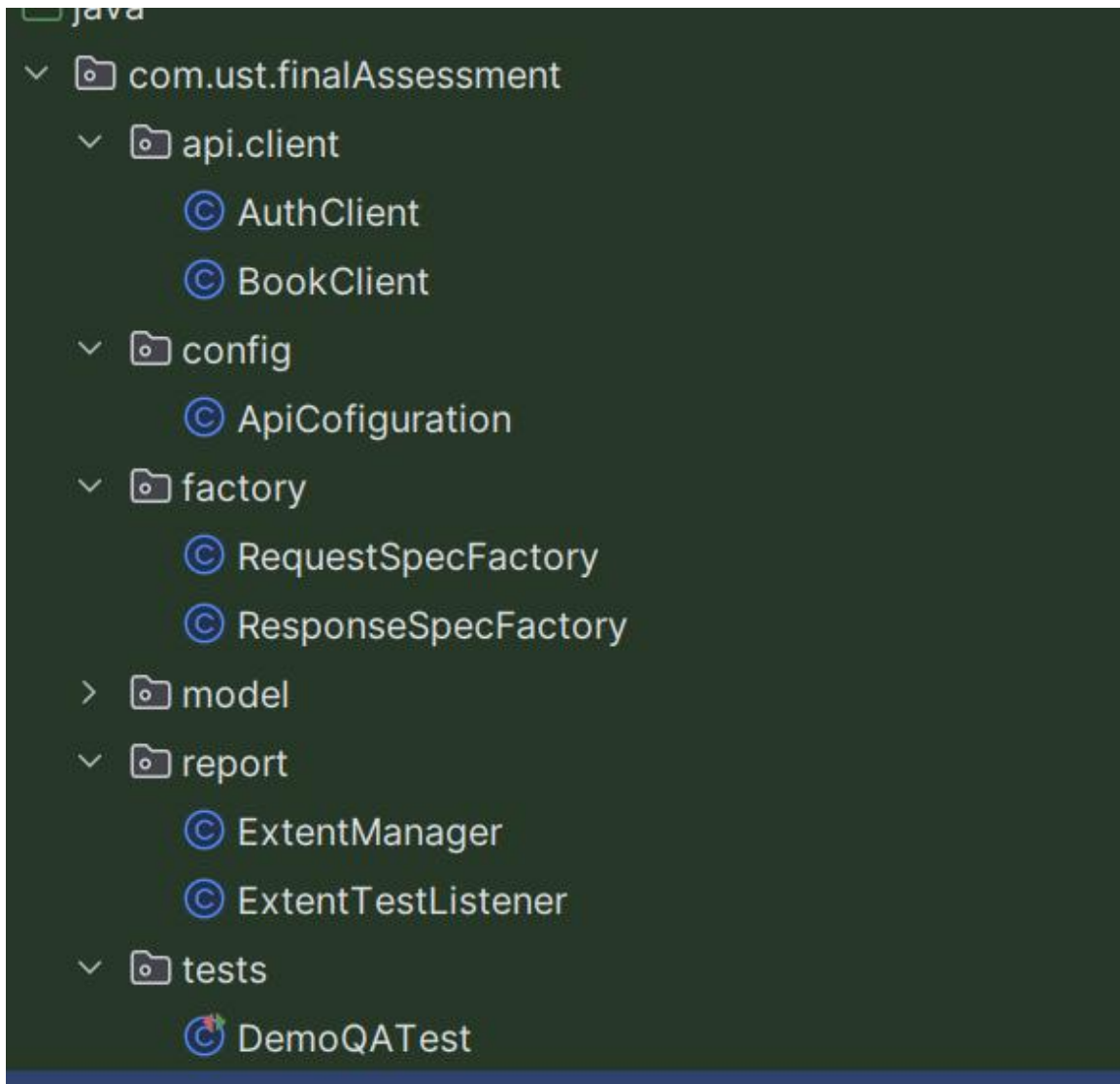
Java, Maven, RestAssured, IntelliJ IDEA / Vs code Allure Reporting.

Workflow

1. Read configuration from Config.java
2. Build request using SpecFactory
3. ClientPage
4. Execute post API
5. Validate response
6. Generate report

Folder Structure





Scenario

Create a user , generate a authentication token, and retrieve this list of books using the token

Commands

- mvn clean test
- mvn test
- allure generate target/allure-results --clean -o target/allure-report

- allure serve target/allure-results

Result Evidence & Screenshots

Code

```
FinalAssessment > SDET-2 > ApiAutomationFinalAssessment > src > test > java > com > ust > finalAssessment > tests > DemoQATest.java > {} com.us
1 package com.ust.finalAssessment.tests;
2 import com.ust.finalAssessment.api.client.AuthClient;
3 import com.ust.finalAssessment.api.client.BookClient;
4 import com.ust.finalAssessment.report.ExtentTestListener;
5 import io.github.cdimascio.dotenv.Dotenv;
6 import io.qameta.allure.*;
7 import io.restassured.path.json.JsonPath;
8 import io.restassured.response.Response;
9 import org.junit.jupiter.api.*;
10 import org.junit.jupiter.api.extension.ExtendWith;
11
12 import static com.ust.finalAssessment.factory.ResponseSpecFactory.*;
13 import static org.hamcrest.MatcherAssert.assertThat;
14 import static org.hamcrest.Matchers.equalTo;
15 import static org.hamcrest.Matchers.greaterThanOrEqualTo;
16
17
18 @Epic("final Assessment Journeys")
19 @Feature("Full Getting Book List")
20 @Owner("Shahbaz Ahmad")
21 @TestMethodOrder(MethodOrderer.OrderAnnotation.class)
22 @ExtendWith(ExtentTestListener.class)
23 public class DemoQATest {
24     static Dotenv dotenv =Dotenv.load();
25     AuthClient authClient =new AuthClient();
26     BookClient bookClient = new BookClient();
27     private final String username = dotenv.get("USER_NAME");
28     private final String password = dotenv.get("USER_PASSWORD");
29     private static String token ;
30     private String id = (Double.toString(Math.random()*1000001));
31 }
```

```
@Test
@Order(0)
@DisplayName("create a user with user name and password")
@Story("user create - generate token - get list of book")
@Severity(SeverityLevel.CRITICAL)
void create_user_with_Credentials()
{
    Response response =authClient.create_User(username+id,password)
        .then()
        .spec(authResponse)
        .extract()
        .response();
    JsonPath jsonPath= response.jsonPath();
    assertThat(jsonPath.getString("username"),equalTo(username+id));
    assertThat(jsonPath.getList("books").size(),equalTo(0));
}
```

```

@Test
@Order(1)
@DisplayName("Generate token")
@Story("user create - generate token - get list of book")
@Severity(SeverityLevel.CRITICAL)
void generate_Token()
{
    Response response =authClient.login_generate_Token(username,password)
        .then()
        .spec(jsonResponse)
        .extract()
        .response();

    JsonPath jsonPath= response.jsonPath();

    token = jsonPath.getString("token");

    assertThat(jsonPath.getString("status"),equalTo("Success"));
    assertThat(jsonPath.getString("result"),equalTo("User authorized successfully."));
}

```

```

@Test
@Order(2)
@DisplayName("Get list of Book by using token")
@Story("user create - generate token - get list of book")
@Severity(SeverityLevel.NORMAL)
void getList_Of_Book()
{
    Response response =bookClient.getListOfBook(token)
        .then()
        .spec(jsonResponse)
        .extract()
        .response();

    JsonPath jsonPath= response.jsonPath();

    var list = jsonPath.getList("books");

    assertThat(list.size(),greaterThanOrEqualTo(0));
}

```

Config.java

```
FinalAssessment > SDET-2 > ApiAutomationFinalAssessment > src > test > java > com > ust > finalAssessment > config > J ApiCofiguration.java > {} com.
1 package com.ust.finalAssessment.config;
2 import io.github.cdimascio.dotenv.Dotenv;
3
4
5 public class ApiCofiguration {
6     static Dotenv dotenv =Dotenv.load();
7
8     public static final String BASE_URL =dotenv.get("BASE_URL");
9 }
10
```

SpecFactory.java

```
FinalAssessment > SDET-2 > ApiAutomationFinalAssessment > src > test > java > com > ust > finalAssessment > factory > J RequestS
1 package com.ust.finalAssessment.factory;
2
3 import io.restassured.builder.RequestSpecBuilder;
4 import io.restassured.specification.RequestSpecification;
5
6 import static com.ust.finalAssessment.config.ApiCofiguration.BASE_URL;
7
8 public class RequestSpecFactory {
9     public static RequestSpecification authRequest =new RequestSpecBuilder()
10         .setBaseUri(BASE_URL)
11         .setBasePath("/Account/v1")//https://demoqa.com/Account/v1/User
12         .addHeader("Content-Type","application/json")
13         .build();
14
15
16     public static RequestSpecification jsonRequest =
17         new RequestSpecBuilder()
18             .setBaseUri(BASE_URL)
19             .setBasePath("/BookStore/v1")
20             .addHeader("Content-Type","application/json")
21             .build();
22 }
```

```

FinalAssessment > SDET-2 > ApiAutomationFinalAssessment > src > test > java > com > ust > finalAssessment > factory > J ResponseS
1  package com.ust.finalAssessment.factory;
2
3  import io.restassured.builder.RequestSpecBuilder;
4  import io.restassured.builder.ResponseSpecBuilder;
5  import io.restassured.http.ContentType;
6  import io.restassured.specification.RequestSpecification;
7  import io.restassured.specification.ResponseSpecification;
8
9  public class ResponseSpecFactory {
10
11      public static ResponseSpecification authResponse =new ResponseSpecBuilder()
12          .expectStatusCode(201)
13          .expectContentType(ContentType.JSON)
14          .build();
15
16
17
18      public static ResponseSpecification jsonResponse =new ResponseSpecBuilder()
19          .expectStatusCode(200)
20          .expectContentType(ContentType.JSON)
21          .build();
22
23
24
25  }
26

```

Result

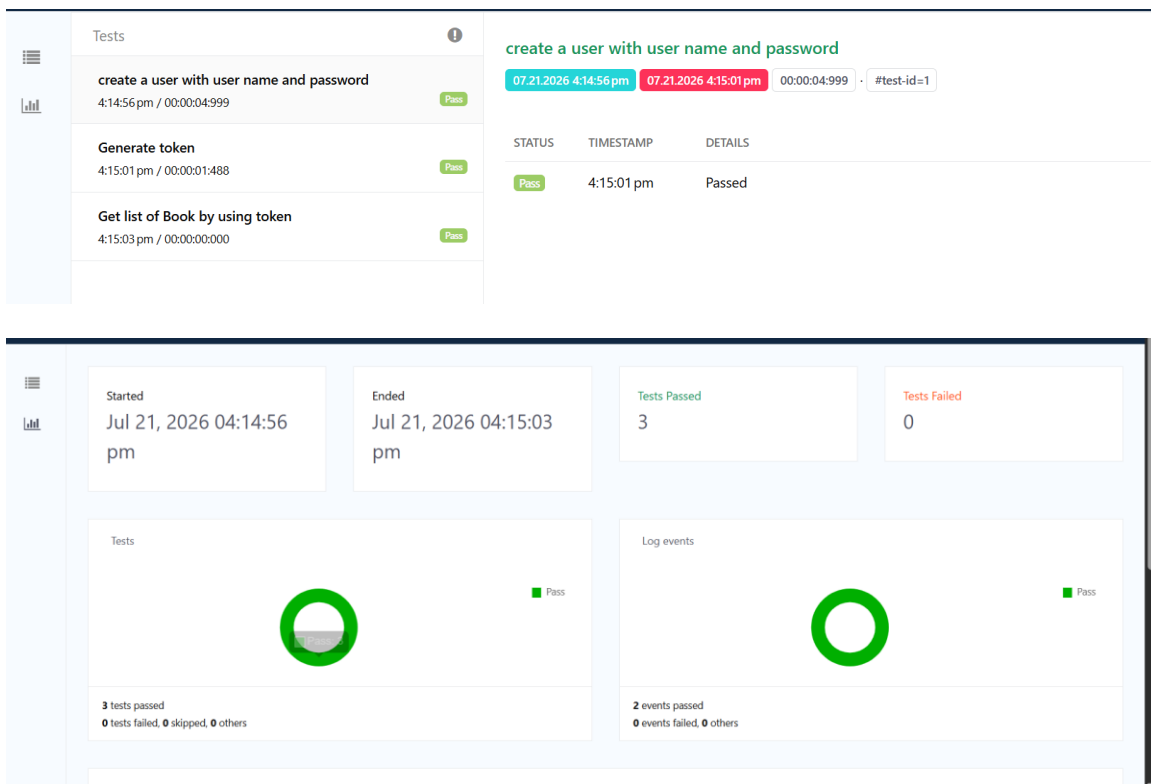
```

[INFO] Finished at: 2020-07-21 10:01:41:00:00
[INFO] -----
PS C:\Users\307424\Documents\QA Training\FinalAssessment\SDET-2\ApiAutomationFinalAssessment> mvn test
[INFO] Scanning for projects...
[INFO] -----< org.example:ApiAutomationFinalAssessment >-----
[INFO] Building ApiAutomationFinalAssessment 1.0-SNAPSHOT
[INFO] from pom.xml
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- resources:3.3.1:resources (default-resources) @ ApiAutomationFinalAssessment ---
[INFO] Copying 0 resource from src\main\resources to target\classes
[INFO] --- compiler:3.13.0:compile (default-compile) @ ApiAutomationFinalAssessment ---
[INFO] Nothing to compile - all classes are up to date.
[INFO] --- resources:3.3.1:testResources (default-testResources) @ ApiAutomationFinalAssessment ---
[INFO] Copying 9 resources from src\test\resources to target\test-classes
[INFO] --- compiler:3.13.0:testCompile (default-testCompile) @ ApiAutomationFinalAssessment ---
[INFO] Recompiling the module because of changed source code.
[INFO] Compiling 8 source files with javac [debug target 22] to target\test-classes
[INFO] Annotation processing is enabled because one or more processors were found
on the class path. A future release of javac may disable annotation processing
unless at least one processor is specified by name (-processor), or a search
path is specified (-processor-path, --processor-module-path), or annotation
processing is enabled explicitly (-proc:only, -proc:full).
Use -Xlint:-options to suppress this message.
Use -proc:none to disable annotation processing.

```

```
Use -proc:none to disable annotation processing.
[INFO] --- surefire:3.5.6:test (default-test) @ ApiAutomationFinalAssessment ---
[INFO] Using auto detected provider org.apache.maven.surefire.junitplatform.JUnitPlatformProvider
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running com.ust.finaAssessment.tests.DemoQATest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 7.202 s -- in com.ust.finaAssessment.tests.DemoQATest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 18.866 s
[INFO] Finished at: 2026-07-21T16:15:04+05:30
[INFO] -----
PS C:\Users\307424\Documents\QA Training\FinalAssessment\SDET-2\ApiAutomationFinalAssessment>
```

Spark Report



Explanation

- First create user by using username and password
- By using same username and password generate authorization token

- By using Authorization token get list of books

Limitations

Dependent on environment configuration, external API availability.

Conclusion

The framework integrates **RestAssured** for robust API testing, enabling validation of backend services and endpoints.

It supports data-driven and reusable API utilities, improving scalability and reducing duplication.

API layers are modularized, ensuring maintainable request/response handling and validation logic.

Enables end-to-end validation, where data created via API can be reused for UI validation.

Ensures high reliability and coverage of backend workflows.

Easily scalable—new API test cases can be added without impacting existing ones.